

Bored, distracted, and forgetful: The impact of mind wandering and boredom on memory encoding

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Abstract

Mind wandering, defined as focusing attention towards task-unrelated thoughts, is a common mental state known to impair memory encoding. This phenomenon is closely linked to boredom. Very few studies, however, have tested the potential impact of boredom on memory encoding. Thus, the present study aimed at manipulating mind wandering and boredom during an incidental memory encoding task, to test their differential impact on memory encoding. Thirty-two participants performed a variant of the *n*-back task in which they had to indicate if the current on-screen object was the same as the previous one (1-back; low working memory load) or the one presented three trials before (3-back; high working memory load). Moreover, thought probes assessing either mind wandering or boredom were randomly presented. Afterwards, a surprise recognition task was delivered. Results showed that mind wandering and boredom were highly correlated, and both decreased in the high working memory load condition, while memory performance increased. Although both boredom and mind wandering predicted memory performance taken separately, we found that mind wandering was the only reliable predictor of memory performance when controlling for boredom and working memory load. Model comparisons also revealed that a model with boredom only was outperformed by a model with mind wandering only and a model with both mind wandering and boredom, suggesting that the predictive contribution of boredom in the complete model is minimal. The present results confirm the high correlation between mind wandering and boredom and suggest that the hindering effect of boredom on memory is subordinate to the effect of mind wandering.

Keywords

Attention; mind wandering; boredom; episodic memory

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Introduction

Mind wandering, most commonly defined as a shift of attention from a task at hand to task-unrelated thoughts (Smallwood & Schooler, 2006), is a fairly common phenomenon estimated to occupy up to 45% of our daily mental activity (Killingsworth & Gilbert, 2010). This high prevalence of mind wandering has a direct impact on the functioning of cognitive processes. Indeed, while it has been pointed out that mind wandering can lead to some positive outcomes, such as an improvement in future thinking (Baird et al., 2011), it is generally associated with a perceptual decoupling of attention leading to a worsened processing of external stimuli (Barron et al., 2011; Schooler et al., 2011; Smallwood et al., 2011). This divided attentional state leads to various negative consequences such as a worsened mood

(Killingsworth & Gilbert, 2010), disruption of sustained attention (McVay & Kane, 2009, 2012; Stawarczyk et al., 2011) and notably a hindered memory encoding. For instance, mind wandering when attending to a lecture course leads to poorer performance when tested afterwards (Hollis & Was, 2016; Wammes, Seli, et al., 2016). Similarly, inductive learning is impaired when experiencing mind

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wandering (Metcalf & Xu, 2016). However, memory is not only mobilised during learning, but is a cognitive function used daily. One of the most prevalent memory systems is episodic memory, which corresponds to the retrieving of information alongside its spatio-temporal context, with a feeling of reviviscence (Tulving, 1972, 2002). Recognition paradigms are often used to assess this memory system and recognition performance was found to be negatively correlated with both self-reported mind wandering (Blondé et al., 2020; Franklin et al., 2016; Hollis & Was, 2016; Maillet et al., 2017; Maillet & Schacter, 2016; Seli, Wammes, et al., 2016; Smallwood et al., 2007; Thomson et al., 2014) and implicit correlates of mind wandering (deBettencourt et al., 2018; Smallwood et al., 2007). Memory performance was also found to be different when comparing participants from high and low mind wandering groups (Riby et al., 2008; Risko et al., 2012). There are several factors which lead to engaging in mind wandering, such as dispositional traits (Andrews-Hanna et al., 2013), task properties (Seli et al., 2018) or negative mood (Smallwood & O'Connor, 2011). Among those negative moods, boredom seems to be predominantly linked to mind wandering.

Boredom can be viewed as the aversive emotional state resulting from the failure to be engaged in a satisfying activity (Eastwood et al., 2012). For instance, when students attend a course on a topic they consider uninteresting, they may fail to focus their attention on the lecture as it provides little excitement to them and thus may experience boredom. This common phenomenon is usually experienced as a transient state disappearing when one's attention is captured again by some stimuli of interest. Boredom's frequency varies among individuals, depending on their boredom proneness (Farmer & Sundberg, 1986; Harris, 2000). Trait boredom is often associated with a worsened mood (Farmer & Sundberg, 1986; LePera, 2011; Vodanovich et al., 1991), while both trait and state boredom are associated with failures of sustained attention (Hitchcock et al., 1999; Hunter & Eastwood, 2018; Malkovsky et al., 2012; Pattyn et al., 2008; Thackray et al., 1977). It is thus not surprising that correlations between boredom and mind wandering have often been reported (Carriere et al., 2008; Critcher & Gilovich, 2010; Danckert et al., 2018; Isacescu et al., 2017; Kane et al., 2007; Steinberger et al., 2016). Mind wandering has been suggested to reduce the dissatisfaction associated with boredom by directing one's attention from un-arousing external stimuli towards more rewarding internal thoughts (Malkovsky et al., 2012; Shepherd, 2019). This corpus of evidence linking boredom and mind wandering matches the subjective phenomenology of mind wandering: when one catches themselves being in mind wandering, it is often due to a lack of interest towards the current occupation (for instance, reading a boring paper, being at the bus stop or in a supermarket queue).

Mind wandering and boredom also seem to be associated with a similar pattern of physiological modifications. Indeed, engaging in task-unrelated thoughts appears to be consistently tied with a short-term elevation in mean heart rate (HR; Cheetham et al., 2016; Ottaviani & Couyoumdjian, 2013; Pham & Wang, 2015; Smallwood, Davies, et al., 2004; Smallwood et al., 2007; Smallwood, O'Connor, et al., 2004). This elevation is considered to be a marker for emotional arousal supposedly generated by the personal salience of internal thoughts (Klinger, 1999). However, this marker may not be entirely reliable, as two studies show no differences in heart rate between mind wandering and being on-task (Ottaviani et al., 2013, 2015). A few studies also pointed out that mind wandering could be associated with a lowered level of skin conductance response (SCR; Blanchard et al., 2014; Cheetham et al., 2016; Smallwood, Davies, et al., 2004; Smallwood et al., 2007). Concerning boredom, a similar pattern of increased HR (London et al., 1972; Merrifield & Danckert, 2014; Pattyn et al., 2008) and decreased SCR (Giakoumis et al., 2010; London et al., 1972; Merrifield & Danckert, 2014) has been reported.

One area where information concerning boredom is rather scarce, compared with mind wandering, is its impact on memory. While boredom is known to be negatively associated with academic performance (Fritea & Fritea, 2013; Pekrun et al., 2010, 2014; Tze et al., 2016) and traits of cognitive failures (Carriere et al., 2008; Wallace et al., 2002, 2003), few studies have tested its direct impact on the encoding process itself. To our knowledge, only two studies (Goldberg & Todman, 2018; Hunter & Eastwood, 2020) tested the impact of boredom on the encoding of either text material or word lists. Taken together, their results suggest that boredom does not have an effect on memorisation itself, but on the valence of the recalled material (Goldberg & Todman, 2018) or that its effect is time-dependent, starting impacting memory after several study sessions (Hunter & Eastwood, 2020). When considering that boredom and mind wandering are two highly intertwined phenomena, these results are quite surprising because a large literature suggests a hindering effect of mind wandering on memory. For instance, similar to boredom, mind wandering is especially prevalent and pervasive during learning (Hollis & Was, 2016; Seli, Wammes, et al., 2016). Furthermore, as previously described, experimental studies have consistently highlighted an impairment of memory encoding due to mind wandering (Blondé et al., 2020; deBettencourt et al., 2018; Seibert & Ellis, 1991; Smallwood, Davies, et al., 2004; Smallwood et al., 2003). Thus, if truly mind wandering impacts memory encoding, and if boredom is so correlated to mind wandering, then we should observe a negative effect of boredom on memory encoding explained by the co-occurrence of mind wandering. Some methodological choices can explain why the two aforementioned studies did not find

an impact of boredom on memory. In the study of Hunter and Eastwood (2020), the induction of boredom took place before the encoding task. As such, it is possible that the associated boredom state did not extend to the task afterwards, explaining the lack of effect. Goldberg and Todman (2018) avoided that problem, as the task itself induced either a low or a high boredom state. In their research, however, the two conditions required a different processing of the information (either reading a short novel or counting vowels in different texts), which may have influenced the results. Goldberg and Todman (2018) have stated, nonetheless, that boredom could affect the encoding process because of its association with mind wandering. However, as no mind wandering assessment was present in their study, this hypothesis remains unchecked, to our knowledge.

One way to test this hypothesis is to assess both boredom and mind wandering levels online during an encoding task and use these measures to predict future retrieval. As boredom reliably arises during monotonous tasks and is associated with a failure to engage the attentional executive control networks (Danckert & Merrifield, 2018), using such a monotonous task would be a straightforward way to induce both boredom and mind wandering. Then, manipulating the cognitive load of the task would be a reliable way to modulate their level (Krimsky et al., 2017; Rummel & Boywitt, 2014; Seli et al., 2018) instead of relying on their spontaneous evolution. Thus, the purpose of this study is to test the impact of both boredom and mind wandering on episodic memory encoding, by manipulating their respective levels. Episodic memory will be assessed through the retrieval of both the item and its encoding context. We postulate that boredom and mind wandering levels will predict one another and will decrease jointly in a high load task, compared to a less engaging task. Concerning mind wandering, there is strong evidence in the literature for predicting a negative relationship with memory performance. Concerning boredom, we predict that a negative relationship with memory will be observed, due to its high correlation with mind wandering.

Material and methods

Participants

Thirty-nine students of the University of Paris (82.86% women, mean age = 19.83 ± 2.25 years) voluntarily attended this study. Recruitment took place from March to May 2019. They signed a declaration of informed consent and were granted a credit necessary to complete their course. Four participants needed to be removed from the analyses due to missing or corrupted data. As mood disorders can have an influence on attention, boredom and memory, three participants were excluded from the analyses as their scores were above two standard deviations of

the Patient Health Questionnaire-4 (PHQ-4) and Satisfaction With Life Scale (SWLS) (see Questionnaires section for more information). The final sample comprised 32 participants (84.38% women, mean age = 19.63 ± 2.22 years). This sample size was deemed sufficient as it is in range with the one used in several studies providing key findings on the effect of mind wandering on memory literature (deBettencourt et al., 2018; Maillet & Rajah, 2014a; Riby et al., 2008; Smallwood et al., 2003; Smallwood, O'Connor, et al., 2004).

Questionnaires

To control the presence of mood disorders, we included the PHQ-4 (Kroenke et al., 2009; Löwe et al., 2010), a very brief questionnaire useful for detecting cues of both depression and anxiety. We also added the SWLS (Blais et al., 1989; Diener et al., 1985) to complement the previous questionnaire with a trait measure. Scoring above two standard deviations on the two PHQ-4 subscales and the SWLS scales was used as an exclusion criterion.

Encoding task

Like in several other studies (Blondé et al., 2020; Chinchanchokchai et al., 2020; Maillet et al., 2017; Maillet & Schacter, 2016), participants had to perform an incidental encoding task. This was done to better represent how memories are formed in natural situations, because most of our memories are not encoded voluntarily. An incident encoding also ensure that participants rely on episodic memory system rather than semantic memory system (Pause et al., 2013). Moreover, as the main paradigm included a cognitive load manipulation, a voluntary encoding would have put the participants in a situation of double task, which would have parasitised the emergence of mind wandering and boredom as well as their influence on memory performance.

The task included two counterbalanced successive blocks both performed by all the participants. Each block was composed of 48 trials presented in a random order. In each trial, one item (the image of a common object such as a tool, a piece of furniture, or food) was presented on-screen. Of these 48 trials, 42 presented different items, while the remaining 6 presented 6 previously seen items which were randomly selected for each participant. The items moved slowly (at a speed of about 3.5 cm/s) from left to right alongside a black line in the centre of the screen, as if they were moving along a factory line (see Figure 1). To reinforce this impression—and to serve as a mask—the image was not visible at first, hidden by a cardboard box image, until it reached the first tier of the line (after about 3 s), at which point the box disappeared and revealed the item. This was done to avoid the processing of the item before it was fully visible on the screen. Two parallel lines

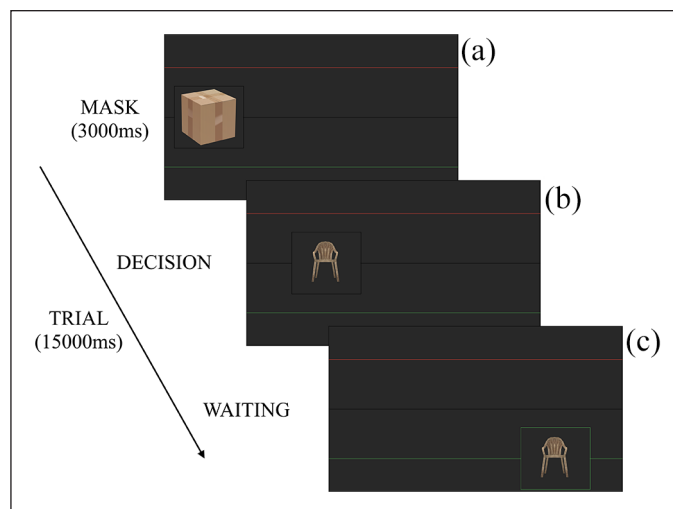


Figure 1. Schematic representation of a trial. The item drifted slowly from left to right alongside the black line, first masked as a cardboard box for the first 3 s of the trial (a). Then, when it appeared, the participant had to redirect it to one of the coloured lines (b). Afterwards, the item could not be moved until the end of the trial, for a total duration of 15 s (c).

were also present on the screen: a red and a green one, respectively, above and below the central line. Both were equidistant from the black line (7.5° of visual angle, at a distance of 60 cm). Participants were informed that they had to perform a classification task on the objects and were told the task was like “being a factory worker processing items on a line.” More precisely, they had to perform an adaptation of the *n*-back task, where they had to compare the image presented on the screen to the previous image (1-back, low working memory load condition) or the image presented three trials before (3-back, high working memory load condition). If the two images matched, participants had to move the item, as quickly as possible, on the red line, with the up key. If not, the image had to be sent on the green line, with the down key. After the participant’s response, the images stayed on-screen until the end of the trial (for a total of about 15 s). Each block corresponded to a single level of cognitive load. Before each block, participants performed a familiarisation block comprising eight geometrical forms to check if they understood the task. Participants were also informed that, from time to time, a thought probe would appear on the screen and temporarily stop the task. Two types of thought probes were presented: mind wandering probes and boredom probes. In the former, participants had to answer by rating their attentional level on a 10-point Likert-type scale ranging from “I’m totally focused on the task” to “I’m totally focused on my thoughts.” In the latter, they had to perform the same task, but on a scale ranging from “The task does not bore me at all” to “The task bores me a lot.” Twelve probes were presented in each block (six mind wandering and six boredom probes), one at a time. Probe types were alternated pseudo-randomly: each probe was paired with another one from a different type. This way, one probe

type could only be presented twice before switching to the other probe type. Alternating probes instead of presenting two probes at once was done to limit the potential disruption caused by two probes appearing at the same point. Probes were dispatched pseudo-randomly during each block, every two to four trials (30–60 s) and stayed on-screen until the participant’s response. Each block lasted for about 15 min.

Recognition task

Memory of previously seen objects was assessed with a recognition task. The 84 images shown in the 2 encoding conditions were presented, alongside 42 new items, in a random order. The presentation was self-paced: images stayed on-screen until the participants’ response. They had three options to answer: “yes” (they were sure the object was presented), “no” (they were sure the object was not presented), and “not sure” (they did not know or were not certain of their answer). If they answered “yes” or “not sure,” they had to answer a context memory question by specifying if the item was presented in the low or the high working memory load condition.

Physiological measures

To provide a thorough assessment of mind wandering and boredom during the task, we complemented the subjective reports with two implicit measures: electrocardiography (ECG) and electrodermal activity (EDA). All physiological measures were recorded using a Biopac MP150 system (Biopac System Inc., USA) alongside the AcqKnowledge Software 4.3 (sampling rate of 1,000 Hz). ECG was recorded using a modified Lead II configuration

Table 1. Mean and standard deviations for the main outcomes associated with the two encoding conditions.

Working memory load	Correct responses (%)	Reaction time (ms)	Mind wandering score	Boredom score	Item recognition (%)	Context recognition (%)	Mean heart rate	Number of SCR peaks
Low	97.52 (5.45)	1,517.34 (629.63)	4.88 (2.62)	5.94 (2.4)	65.71 (20.06)	62.30 (21.25)	82.63 (9.39)	1.01 (0.76)
High	94.5 (10.43)	1,903.64 (966.86)	2.79 (2.53)	3.73 (2.67)	75.43 (14.02)	77.07 (14.72)	85 (9.5)	0.93 (0.86)

SCR: skin conductance response.

(Makowski, Sperduti, Pelletier, et al., 2019; Takuma et al., 1995), by placing three electrodes (EL504) on the torso: one on the right sub clavicular space and two on the left and right lower ribs. Finally, EDA was recorded with two Ag/AgCl electrodes (EL507) placed on the intermediate phalange of the index and middle fingers of the left hand.

Procedure

The experiment was conducted in a quiet room of the Institute of Psychology at the University of Paris. Participants sat in a chair at approximately 60 cm of a 22.9" computer screen. They started the experiment by filling a personal information form and signing an informed consent. Electrodes were put in place during this part, and a test recording was performed to check if the signal was recorded correctly. If not, the electrodes were put in place once more until the signal was clean. Afterwards, participants completed the first block of the encoding task, presented as an attentional task. The order of the two blocks was counterbalanced for each participant. Before each block, participants completed a training task comprising eight stimuli (simple geometrical forms, one being repeated twice) and a thought probe at the end, to ensure they understood the instructions before starting the main task. Then, acting as an interference task between encoding and recognition, participants completed the two questionnaires and participated to a brief cognitive control evaluation. This evaluation included a simple reaction time task, a choice reaction time task, a Go-No Go task, and a flanker task. The detailed description of the task is reported elsewhere (Makowski, Sperduti, Blondé, et al., 2019). The retention interval lasted for about 15 min. Finally, participants performed a surprise recognition task before going through a debriefing. A typical experimental session lasted for about 1½ hr.

Data analysis

Indices computation

First, we have calculated the percentage of correct responses during the encoding task, by dividing the number of hits (when participants correctly determined if the item was previously seen or not) by the total number of trials. Then, we calculated the percentage of correctly recognised items, based on the number of correct "yes" answer to the total number of images presented once

(excluding the "not sure" answers and the repeated images). Then, using the same criteria, we have calculated the percentage of correctly recognised context, based on the number of correctly retrieved context condition to the total number of correct "yes" responses.

For the physiological measures, we have calculated the mean heart rate and the number of SCR peaks for each trial preceding a probe over a 15-s window before the probe's appearance. This window was selected instead of longer epoch durations used in similar protocols (Ottaviani et al., 2013; Smallwood, Davies, et al., 2004; Smallwood et al., 2007; Smallwood, O'Connor, et al., 2004) to prevent the overlapping of several trials. The ECG signal was filtered with a high-pass finite impulse response (FIR) filter (with a cutoff frequency of 1 Hz), and the EDA signal was filtered with a low-pass FIR filter (with a cutoff frequency of 0.3 Hz), in line with the current literature (Blanchard et al., 2014; Smallwood, Davies, et al., 2004; Smallwood et al., 2007; Smallwood, O'Connor, et al., 2004). The SCR peaks were detected based on a detection threshold of 0.5 standard deviations. For a descriptive summary of each variable, see Table 1.

Analysis methods

R 3.6.0 (R Development Core Team, 2008) was used to conduct all statistical analyses. In line with recent statistical guidelines (Etz & Vandekerckhove, 2016; Wagenmakers et al., 2018), we used the Bayesian framework to analyse our data, using the packages *rstanarm* (Carpenter et al., 2017; Gabry & Goodrich, 2016) and *psycho* (Makowski, 2018). In the following analyses, we will express our results in terms of certainty levels. An effect will be considered as likely existing (similar to $p < .05$) if its *pd* (probability of direction, a Bayesian indicator of the existence of an effect) is above 97% (Makowski, Ben-Shachar, et al., 2019). Indicators are associated with a 90% Credible Interval (CI), deemed more appropriate for Bayesian analyses than the 95% interval usually used in the frequentist approach (Kruschke, 2014). For model comparisons, we used the leave-one-out (LOO) cross-validation method to assess which model has the best predictive power (Gelman et al., 2014; Vehtari et al., 2017). For each model, we reported two indicators of model performance: the expected log predictive density (ELPD) and the

leave-one-out information criterion (LOOIC). The ELPD assesses the posterior predictive distribution of the model while the LOOIC is an estimation of ELPD using another dataset (Gelman et al., 2014; Vehtari et al., 2017). The higher the ELPD is—and the lower the LOOIC is—the better the model's predictive power will be. Finally, Bayes factors (BFs) were computed to pit each model against the others. BF estimation quantifies the support for a hypothesis against another. As such, the more the BF is greater than one, the higher the models are likely to differ. Conversely, the more the BF is lower than one, the higher the models are likely *not* to differ (Kass & Raftery, 1995; van Doorn et al., 2020). All models included both the encoding condition and the order of the conditions as fixed effects. All logistic models predicting memory performance did not include repeated items or “not sure” responses. Furthermore, all our models included the participants as random effect. Additional fixed and random effects are stated when necessary. The analyses were either performed at the participant level (testing the prediction of the participants' mean scores) or at the item level (testing the prediction of each item). To help clarify the distinction between the two levels of analyses, this information was specified each time a model is described. For more information, each model is reported in the supplementary material.

Results

Manipulation check

First, we have tested if the encoding task was correctly done by our participants. Thus, we have computed a participant-level mixed Bayesian linear model predicting the percentage of correct responses with the working memory load of the encoding condition (with item occurrence as fixed effect). The difference in the percentage of correct responses between the low and high working memory load conditions had a 98.28% probability of being negative (Median = -2.82, 90% CI = [-4.89, -0.58], $R^2 = .23$). This suggests it is likely that working memory load affected the percentage of correct response, which was better in the low memory load condition ($97.52\% \pm 5.45\%$) than in the high memory load condition ($94.5\% \pm 10.43\%$). Furthermore, to complement the previous analysis, we have computed an item-level mixed Bayesian linear model predicting the reaction times of each trial with the corresponding working memory load (with item occurrence as fixed effect, and trial number and stimulus as additional random effects). For the reaction times to be comparable, we have converted the raw data into within-participant z scores. The difference in reaction times between the two conditions had a 100% probability of being positive (Median = 0.58, 90% CI = [0.5, 0.66], $R^2 = .11$). Thus, it is almost certain that working memory load also had an effect on reaction times, which were shorter in the low memory load condition (z -scores: -0.27 ± 0.8 ; raw scores:

$1,517.34 \pm 629.63$ ms) than in the high memory load condition (z -scores: 0.28 ± 1.1 ; raw scores: $1,903.64 \pm 966.86$ ms).

Then, we tested if we successfully impacted the occurrence of mind wandering and boredom episodes by manipulating the working memory load of the encoding task. We have computed two participant-level mixed Bayesian linear models to predict mind wandering or boredom scores with the encoding condition. Concerning mind wandering, the difference of mind wandering scores between low and high working memory load conditions had a 100% probability of being negative (Median = -1.99, 90% CI = [-2.3, -1.67], $R^2 = .6$), meaning that there is an almost certain effect of the working memory load on mind wandering, with a higher mind wandering score for the low working memory load condition (4.88 ± 2.62) than for the high memory load condition (2.79 ± 2.53). Concerning boredom, the difference of boredom scores between low and high working memory load conditions had a 100% probability of being negative (Median = -2.12, 90% CI = [-2.39, -1.84], $R^2 = .68$). Thus, the working memory load predicts almost certainly boredom, with a higher boredom score when the working memory load is low (5.94 ± 2.4) than when it is high (3.73 ± 2.67). See Figure 2 for a comparison. Finally, a mixed Bayesian linear model was performed to predict the mean mind wandering score with the mean boredom one. The effect of boredom on mind wandering had a 100% probability of being positive (Median = 0.46, 90% CI = [0.28, 0.65], $R^2 = .64$). Thus, it is almost certain that boredom linearly predicts mind wandering.

Finally, we assessed if mind wandering and boredom influenced the accuracy of the working memory task during encoding. To this end, we computed two participant-level mixed Bayesian linear models to predict the percentage of correct responses with either mind wandering or boredom as predictors. The effect of mind wandering had a 71.25% probability of being negative (Median = -0.16, 90% CI = [-0.62, 0.31], $R^2 = .22$) and the effect of boredom had a 77.33% probability of being negative (Median = -0.2, 90% CI = [-0.62, 0.27], $R^2 = .21$). Thus, it is uncertain whether mind wandering and boredom predicted working memory performance.

Prediction of memory with working memory load

We have computed two participant-level mixed Bayesian linear models with the working memory load at encoding as a predictor and either the percentage of correctly recognised items or the percentage of correctly identified context as predicted variables. For item memory, the difference in percentage of item recognition between the two encoding conditions had a 99.93% probability of being positive (Median = 10.24, 90% CI = [5.56, 15.12], $R^2 = .61$). For context memory, the difference in percentage of context recognition between the two conditions had a 99.98%

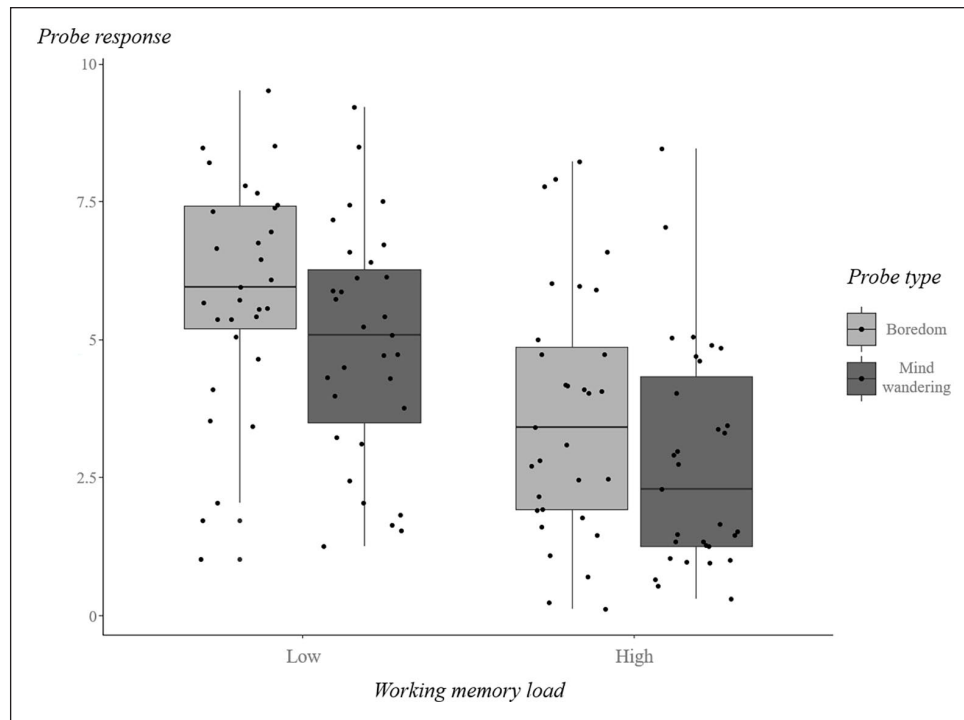


Figure 2. Whisker plot of boredom and mind wandering scores for each condition of working memory load.

probability of being positive (Median=15.24, 90% CI=[7.66, 22.21], $R^2=.25$). Thus, it is very probable that encoding condition impacted memory, as we observed a lower percentage of recognition for both items ($65.71\% \pm 20.06\%$) and their context ($62.30\% \pm 21.25\%$) in the low working memory load condition than the percentage of recognition for items ($75.43\% \pm 14.02\%$) and their context ($77.07\% \pm 14.72\%$) in the high working memory load condition.

As memory performance can be driven by the time spent processing the stimuli, we computed two item-level Bayesian logistic models predicting either the probability of correctly recognising an item or the probability of correctly recognising its context with the reaction time (z-score) associated with the item (with trial number and stimulus as additional random effects). The effect of reaction time on item memory had a probability of 64.8% to be positive (Median=0.03, 90% CI=[-0.08, 0.13], $R^2=.21$), and its effect on context memory had a 97.95% probability of being positive (Median=0.13, 90% CI=[0.03, 0.24], $R^2=.11$). Thus, while it is uncertain whether reaction time can predict item memory, it is likely to predict context memory.

Prediction of memory with mind wandering

To test the impact of mind wandering on later item recognition, we have computed an item-level Bayesian logistic model predicting the probability of correctly

recognising an item placed right before a probe with the score given to that probe (with trial number and stimulus as additional random effects). The linear prediction of a correct recognition by mind wandering score had a 99.98% probability of being negative (Median=-0.30, 90% CI=[-0.45, -0.14], $R^2=.28$). Thus, it is very likely that mind wandering during the encoding of an item negatively predicts its subsequent recognition (see Figure 3).

Then, to test the impact of mind wandering on the correct retrieval of the context condition, we have computed the same item-level Bayesian logistic model (with the addition of z-scored reaction times as fixed effect), with the ability to retrieve the encoding condition as the predicted variable. In this model, the linear relationship between mind wandering and successful context retrieval had a 96.68% chance of being negative (Median=-0.18, 90% CI=[-0.35, 0.01], $R^2=.33$). Thus, mind wandering at encoding may possibly predict the probability of retrieving context condition.

Prediction of memory with boredom

Next, to test the impact of boredom on subsequent item recognition, we have computed an item-level Bayesian logistic model predicting the probability of correctly recognising an item situated before a boredom probe with the given score to that probe (here too with trial number and stimulus as additional random effects). It appears the

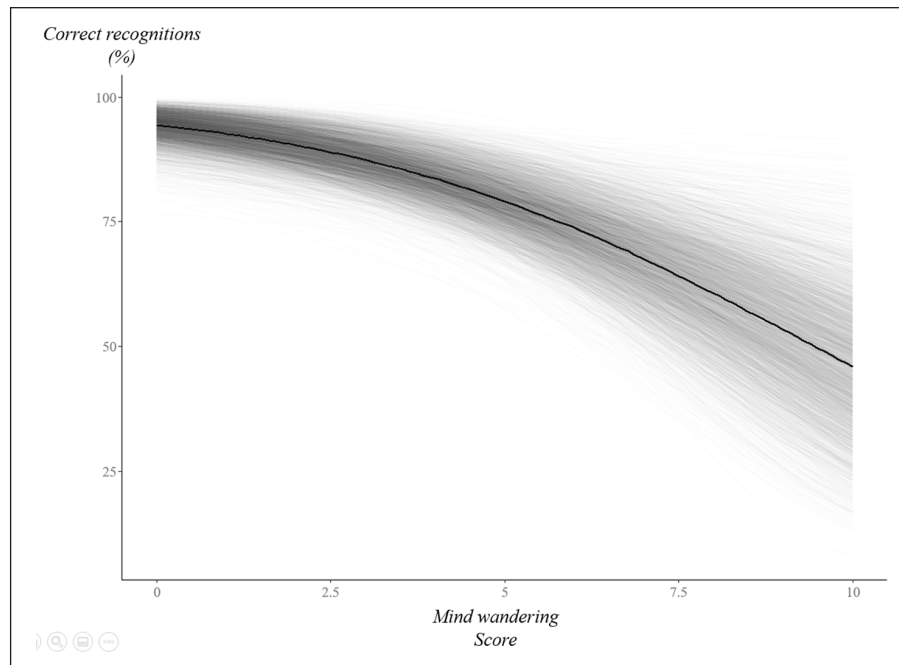


Figure 3. Probability of obtaining a correct item recognition based on the mind wandering score at encoding. The bold line corresponds to the median effect, while the thin lines represent 4,000 possible draws from the posterior distribution.

linear prediction of a correct recognition by boredom score had a 98.38% probability of being negative (Median = -0.19 , 90% CI = $[-0.35, -0.03]$, $R^2 = .30$). As such, experiencing boredom at encoding is likely to negatively predict later item memory performance (see Figure 4).

Then, for context memory, we have computed the same item-level Bayesian logistic model (also with the addition of z-scored reaction times as fixed effect), with the ability to retrieve the encoding condition as the predicted variable. The linear relationship between correctly retrieving the context condition and boredom had a 74.43% chance of being negative (Median = -0.06 , 90% CI = $[-0.21, 0.09]$, $R^2 = .21$). Thus, it is uncertain whether boredom had an impact on context retrieval.

Model comparison for item memory

As mind wandering, boredom, and the working memory load reliably predicted subsequent item memory, we performed additional analyses to determine which one of these predictors was the most reliable.

First, we have computed a single participant-level mixed Bayesian linear model to predict the overall percentage of correctly recognised items with the working memory load, mind wandering, and boredom mean scores as predictors. The model had an explanatory power (R^2) of about .67. Within this model, the effect of mind wandering had a 99.8% probability of being negative (Median = -3.36 , 90% CI = $[-5.10, -1.50]$). On the contrary, the effect of boredom had a 67.13% probability of being negative (Median = -0.46 ,

90% CI = $[-2.03, 1.4]$). Moreover, the effect of working memory load at encoding had a 76.08% chance of being positive (Median = 2.43 , 90% CI = $[-3.29, 8.41]$). Thus, when taking into account all potential predictors, it is very likely that mind wandering negatively predicts item memory performance, while it is uncertain whether boredom and the encoding condition can predict it.

Then, we performed model comparisons to specifically assess the respective contribution of boredom and mind wandering as predictors of item memory. We have compared the previously described model (the complete model) to three reduced model: one without boredom (the mind wandering model), one without mind wandering (the boredom model), and one without both (the base model). Comparing those models using cross-validation showed that the mind wandering model ($d_{\text{ELPD}} = -257.09$, $d_{\text{LOOIC}} = 514.18$) outperformed the complete model ($d_{\text{ELPD}} = 0.81$, $d_{\text{LOOIC}} = 1.62$), the boredom model ($d_{\text{ELPD}} = 5.36$, $d_{\text{LOOIC}} = 10.72$), and the base model ($d_{\text{ELPD}} = 6.05$, $d_{\text{LOOIC}} = 12.1$). As such, a model including only mind wandering appears to be the best way to predict item memory (see Table 2 for an overview of the models).

Finally, to assess how well the data were described by each model relative to the others, we computed BF_s for each pair (see Table 3 for the comparison matrix). It appears that every model had a very substantial chance of describing the data better than the base model, with the complete model having the highest chance (BF = 4,645.51), then the mind wandering model (1,652.44), and finally the boredom one (BF = 18.11). We then used the complete

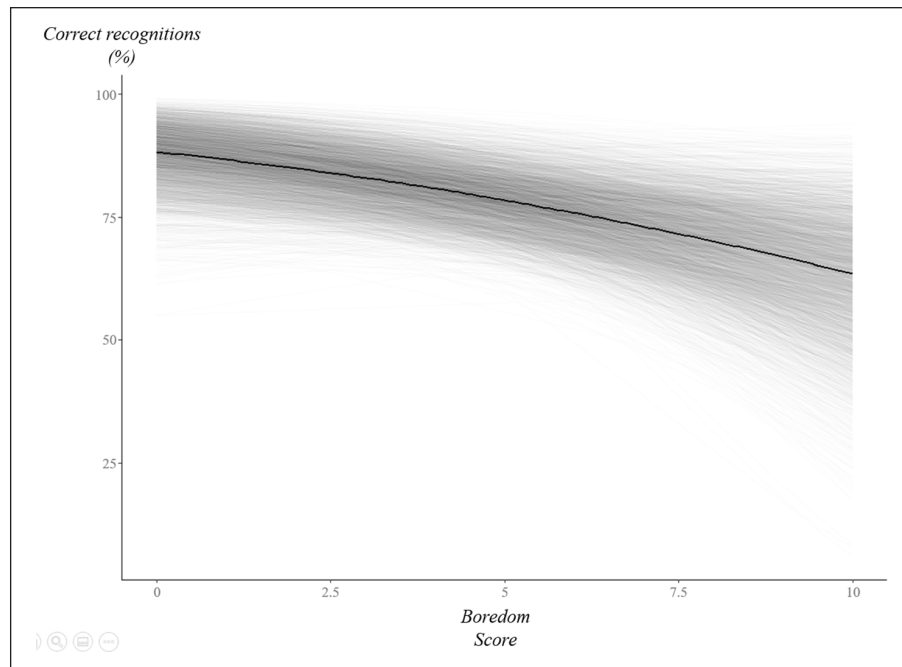


Figure 4. Probability of obtaining a correct item recognition based on the boredom score at encoding. The bold line corresponds to the median effect, while the thin lines represent 4,000 possible draws from the posterior distribution.

Table 2. Comparison of models predicting item memory ranked by performance using the expected log point-wise predictive density (ELPD) and the LOO information criterion (LOOIC).

Models	ELPD	LOOIC	R^2
Mind wandering	-257.09	514.18	.68
Complete model	-257.90	515.80	.67
Boredom	-262.45	524.90	.61
Base model	-263.14	526.28	.61

The best predictors are reported in bold.

Table 3. Comparison matrix of the four models predicting item memory using Bayes factor (BF).

Models	Mind wandering	Boredom	Complete model
Boredom	92.35	—	—
Complete model	0.36	0.004	—
Base model	1,652.44	18.11	4,645.51

Each column corresponds to the model used as a numerator in the calculation of the BF, while the row refers to the denominator. The more the BF is greater than one, the more evidence is provided for a difference in how the data are described by the models. The more the BF is lower than one, the more evidence is provided for no difference in how the data are described.

model as denominator to assess if there was evidence that the mind wandering and boredom models were more informative. There was very strong evidence that the boredom model did not describe the data better than the

complete model ($BF=0.004$) and anecdotal evidence that the mind wandering model did not describe the data better either ($BF=0.36$). Finally, specifying the boredom model as denominator and the mind wandering one as numerator, there was very strong evidence that the mind wandering model described the data better ($BF=92.35$). Thus, based on the BF, both the mind wandering and complete models should be preferred to account for memory performance.

Physiological prediction

Finally, we have computed two item-level mixed Bayesian linear models to predict either the level of mind wandering or boredom using the mean HR measured 15 s before the probe (with the trial number and stimulus as additional random effects). First, heart rate's effect on mind wandering had a 71.48% probability of being negative (Median = -0.01, 90% CI = [-0.04, 0.02], $R^2 = .62$). Second, heart rate's effect on boredom had a 99.5% probability of being negative (Median = -0.05, 90% CI = [-0.09, -0.02], $R^2 = .69$). Thus, heart rate is very likely to negatively predict boredom while it is uncertain whether it can predict mind wandering.

Concerning skin conductance, we have computed the same two previous item-level Bayesian linear models again but using as predictor the number of SCR peaks measured 15 s before the probe. The two models were found to be inconclusive. The effect of the number of peaks on mind wandering had a probability of 80.95% to

be negative (Median = -0.22, 90% CI = [-0.65, 0.18], $R^2 = .52$), and its effect on boredom had a 74.38% probability of being negative (Median = -0.16, 90% CI = [-0.56, 0.24], $R^2 = .67$). Thus, it is uncertain whether SCR can predict either mind wandering or boredom.

Discussion

This study aimed at testing whether boredom could affect episodic memory encoding through the emergence of mind wandering. We manipulated cognitive load at encoding to modulate mind wandering and boredom. Several findings emerged from this study. First, we were successful in manipulating both mind wandering and boredom through a change in working memory load. Second, mind wandering and boredom were, as hypothesised, highly correlated. Third, both mind wandering and boredom negatively predicted item memory performance, but only mind wandering tended to predict context memory. When computed into a single model, mind wandering appeared to be a strong predictor of item memory, while boredom was not a reliable predictor. The model including mind wandering also better explained the results compared to the one including boredom. And finally, while skin conductance was not found to be a predictor of either mind wandering or boredom, a decrease in heart rate was found to predict boredom, but not mind wandering.

The paradigm used in this article was effective at modulating both mind wandering and boredom levels and impacting memory performance through a modification of working memory load during encoding. The manipulation of cognitive load is confirmed by both the lower percentage of correct responses and the increase in reaction time found in the high working memory load condition. The difference in working memory task accuracy was found to be mostly driven by task difficulty because mind wandering and boredom were not found to be reliable predictors. This pattern of results could stem from the fact that, despite the difference between conditions, response accuracy was very high even in the high load condition. This ceiling effect could have prevented from finding an effect of mind wandering and boredom. Also, it should be noted that the reaction times were overall quite long when compared to regular *n*-back tasks (Jaeggi et al., 2010), especially for the low working memory load condition. This is likely due to the design of our modified *n*-back task, as stimuli were presented at a slow-paced rate (one every 15 s) while in classical *n*-back tasks, items are usually immediately presented after one another (Conway et al., 2005). While the instruction was to answer as quickly as possible, it is possible that the slow pace of the experiment did not incentivise fast reactions. Moreover, the most common material used in *n*-back tasks is digits, which are simple stimuli quick to process. Processing more complex material, such as pictures can lead to higher reaction times (Palomäki

et al., 2012). Despite this, we still found noticeably slower reaction times in the 3-back condition compared to the 1-back one, confirming the difference in working memory load, irrespective of the base reaction time. Another finding is that, regardless of the working memory load, mind wandering and boredom were found to be extremely correlated, which is in line with past literature framing the two processes as co-occurrent (Carriere et al., 2008; Critcher & Gilovich, 2010; Danckert et al., 2018; Kane et al., 2007; Steinberger et al., 2016). However, the causality of this association is still unclear (Raffaelli et al., 2018). For instance, some studies have pointed out that mind wandering could be a sort of “defence mechanism” against boredom, triggering when external stimuli are under-arousing, to redirect attention to more appealing internal stimuli (Malkovsky et al., 2012; Shepherd, 2019). The opposite hypothesis has been proposed too: boredom could be the emotional response emerging from the discrepancy between internal interesting stimuli and external uninteresting ones (Eastwood et al., 2012). Dissociating mind wandering and boredom through experimental design should be a step towards a better understanding of their relationship and the specificities of each process.

Apart from the decrease in both mind wandering and boredom, increasing task difficulty was associated with an improvement in episodic memory performance. The better retrieval emerging from the high load encoding condition may be explained in two different ways: either the encoding was better because the nature of the task forced the participants to process the information for a longer time, or it stems from the higher focus on the task due to the lack of mind wandering. The former hypothesis is not supported by our data concerning memory for items, as longer reaction times did not lead to a better encoding, but interestingly this relationship emerged for context memory. Encoding the context of presentation of an item requires the mobilisation of an associative process between the item and its context (Ranganath, 2010). As such, the successful execution of this additional process may be more dependent on processing time than item encoding only, which was more affected by the distribution of attentional resources. Indeed, the effect of task difficulty on item recognition was uncertain when taking into account the level of mind wandering at encoding, which emerged as a strong predictor of memory performance. Thus, while a direct effect of the task cannot be entirely excluded, our results suggest it was more likely that the increase in mind wandering affected item memory performance rather than the task itself. Moreover, the fact that mind wandering during encoding led to a worsened recognition fits into the perceptual decoupling framework. It is likely that participants experienced an attentional shift during mind wandering, prioritising the processing of internal stimuli instead of external ones, thus leading to a poorer encoding. As such, experiencing mind wandering during encoding may be

similar to a divided attention situation, a phenomenon well-known to have a hindering effect on memory encoding (Craig et al., 1996; Curran, 2004; Fernandes & Moscovitch, 2000; Uncapher & Rugg, 2005). If the effect of mind wandering on item memory starts to become well documented, this study is one of the few suggesting a potential hindering effect on context memory as well (Maillet & Rajah, 2014b, 2016), which may highlight a detrimental impact of mind wandering on the binding between the items and their context. As such, the assimilation of mind wandering with divided attention holds true, as dividing attention at encoding also leads to a hindering effect on context memory (Craig et al., 2010; Troyer & Craig, 2000; Troyer et al., 1999). This effect, however, should be relativised, as reaction times were shown to predict context recognition. Hence, the effect of mind wandering may have been reduced by taking into account the effect of reaction time. Also, as context recognition was only assessed for correctly recognised items, the diminished impact of mind wandering may stem from the fact that these items were, on average, associated with lower rates of mind wandering.

Boredom, by contrast, was a slightly less reliable predictor of memory performance, as it only led to a decrease in item memory, with no impact on context memory. This lack of precision may be due to the fact that, according to our hypothesis, the effect of boredom was indirect and was mostly driven by its high correlation with mind wandering. Supporting this hypothesis, we found that, when predicting item memory performance with both mind wandering and boredom, only mind wandering remained a reliable predictor. This can be interpreted as the effect of boredom being subsumed to the effect of mind wandering. This hypothesis is supported by the results of our item memory model comparisons. Each model had a high probability of having a better predictive power than the base model, but when comparing the boredom model to the mind wandering and complete models, substantial evidence showed that the boredom model was outperformed by the mind wandering model and did not describe the data better than the complete model. The mind wandering model was not found to describe the data better than the complete model either, but the evidence was far much weaker. It is thus possible that most of the predictive power of the complete model was attributed to mind wandering, and only marginally to boredom. Furthermore, when observing the ELPD scores, the fact that the complete model—including both mind wandering and boredom—had slightly less predictive power than the mind wandering one suggests that adding boredom as a predictor is redundant. As such, it is fair to assume that the boredom effect on item memory performance is a by-product of the mind wandering effect, due to the high correlation between the two predictors.

The fact that boredom predicted item memory is at odds with previous results showing no direct effect on memory

encoding (Goldberg & Todman, 2018; Hunter & Eastwood, 2020). As mentioned in the introduction, some methodological choices can account for the lack of effect in these previous experiments, namely the fact that boredom was induced before the task, and not by the task itself (Hunter & Eastwood, 2020) and the fact that the two conditions of the task required different processing of the encoded information (Goldberg & Todman, 2018). The task used in this study managed, we believe, to address those two problems by manipulating and assessing the levels of boredom during the task itself, while only changing the quantity of information to process, and not the nature of the task. However, it is also possible that we did not assess the same kind of boredom than these studies. Indeed, Hunter and Eastwood (2020) pointed out the need to differentiate between boredom proneness, state boredom, and task boringness. In the present experiment, the boredom probes prompted the participants to rate how much they considered *the task* to be boring. Hence, the results we found may be more tied to task boringness rather than boredom state per se. However, it is quite counterintuitive that a task perceived as boring does not lead to state boredom, especially when the participant cannot engage in another activity, as it is the case in a laboratory experience. It is possible, though, that engaging in pleasant mind wandering may lead to not feeling bored while still perceiving the task as boring. More research would be needed to see whether such dissociation could be observable. Finally, assessing the phenomenology of boredom could also help to understand more precisely the relationship between boredom, mind wandering, and memory. Indeed, boredom can be viewed in two ways: either a low arousal state (“apathetic boredom”) characterised by a feeling of disinterest (Mikulas & Vodanovich, 1993; Pattyn et al., 2008; Vogel-Walcutt et al., 2012), or a high arousal state (“agitated boredom”) associated with a feeling of restlessness (Jang et al., 2015; Martin et al., 2006; Merrifield & Danckert, 2014). While mind wandering is consistently associated with boredom, it was also found to be specifically associated with restlessness, but not sleepiness (Danckert et al., 2018). It might thus be possible that only agitated boredom leads to memory impairment, due to its ties to mind wandering.

In contrast to our behavioural results, boredom and mind wandering did not present a similar pattern of physiological measures. Indeed, boredom appeared to be reliably predicted by heart rate, while mind wandering was not, and skin conductance variations predicted neither of these two mental states. Surprisingly, the relationship between heart rate and boredom was negative, while an association with an increased heart rate is a consistent finding in previous literature (London et al., 1972; Merrifield & Danckert, 2014; Pattyn et al., 2008). This finding may stem from our experimental design, as some studies pointed out that the change in heart rate was not associated with boredom itself but rather a change in working memory load (Hill &

Perkins, 1985; Perkins, 1981). Also, in contrast to most studies, mind wandering was not reliably predicted by either heart rate or skin conductance. However, these results should be taken with great caution as they may be derived from experimental noise. Indeed, the relative number of probes (six for each type) coupled with the rather small sample size means that few epochs were recorded, which can partially explain our lack of results. Nevertheless, it should be noted that some studies also failed to highlight a relationship between mind wandering and heart rate (Ottaviani et al., 2013, 2015).

One limitation to this study is our inability to determine which kind of mind wandering participants experienced. Indeed, while generally considered a disruptive phenomenon, the effect of mind wandering on encoding may be modulated by different characteristics. For instance, some authors propose to distinguish between spontaneous and deliberate forms of mind wandering (Robison & Unsworth, 2017; Seli et al., 2015; Seli, Risko, & Smilek, 2016). The former corresponds to a natural, uncontrolled switch from on-task to off-task thoughts while the latter represents a conscious choice of voluntarily focusing on off-task thoughts. It has been suggested that these two forms differently affect memory performance in academic learning. By testing memory performance after each lecture and at the end of a semester, spontaneous mind wandering was found to hinder final test performance, while deliberate mind wandering was negatively correlated with post-lecture test performance (Wammes, Boucher, et al., 2016; Wammes, Seli, et al., 2016). Moreover, Isacescu et al. (2017) have shown that boredom proneness was significantly predicted by spontaneous mind wandering but not by deliberate mind wandering. Given the strong correlation between boredom and mind wandering in this study, we can assume that our participants mostly experienced spontaneous mind wandering. Making such a distinction would have required to include additional thought probes focused on the nature of mind wandering. As a matter of fact, some studies used extensive probing methods to cover various aspects of mind wandering's content (Hollis & Was, 2016; Kane et al., 2017; Maillet et al., 2017; Maillet & Schacter, 2016). However, this type of assessment would not have fit well with our encoding paradigm. It is especially true for the high working memory load condition, as it would have greatly disrupted the maintenance of items in working memory.

Overall, this study both highlights a potential negative effect of boredom on memory performance and suggests that this effect may be tied to mind wandering accompanying this emotional state. The next step in understanding the respective influence of these two phenomena on episodic memory would be to create an experimental dissociation between the two. Manipulating task difficulty proved to be unsuccessful in the present experiment, potentially because task engagement was stronger in the high working memory

load condition. One potential avenue to achieve this dissociation could be the manipulation of the expertise of the participants. Indeed, some automatised activities can help reducing boredom while leaving attentional resources available to engage in mind wandering. For instance, playing video games could provide such a situation. Indeed, this activity can be considered enjoyable enough to prevent—or limit—the emergence of boredom, while being automatised through expertise, allowing the emergence of mind wandering for expert players. As such, when playing a video game, expert and novice players may differ in their mind wandering levels while maintaining similar levels of boredom. This dissociation could then be used to test the respective influence of mind wandering and boredom for memory performance of events appearing in the video game.

Conclusion

This study represents a novel effort to better assess and understand the respective effects of mind wandering and boredom on episodic memory encoding. By manipulating mind wandering and boredom through an increase in working memory load at encoding, we found that both boredom and mind wandering decreased and predicted later item recognition, while only mind wandering tended to predict context retrieval. However, when conducting more advanced analyses, mind wandering appeared to be the best predictor of item memory, while boredom's effect became uncertain. Mind wandering and boredom, often considered as two faces of the same coin, need further research to understand more precisely their relationship and better delineate their respective impacts on cognitive functioning.

Availability of data and material

The present study was not pre-registered. The raw data and scripts for their processing and statistical analyses are available in the following depository: <https://osf.io/87tnw/files/>.

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Ethical standards

The manuscript does not contain clinical studies or patient data. All participants signed an informed consent before attended to this experiment.

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Supplementary material

The supplementary material is available at qjep.sagepub.com.

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